

Part 1 — Architecture Explanation

Data Flow: From Operational Database to Business Intelligence

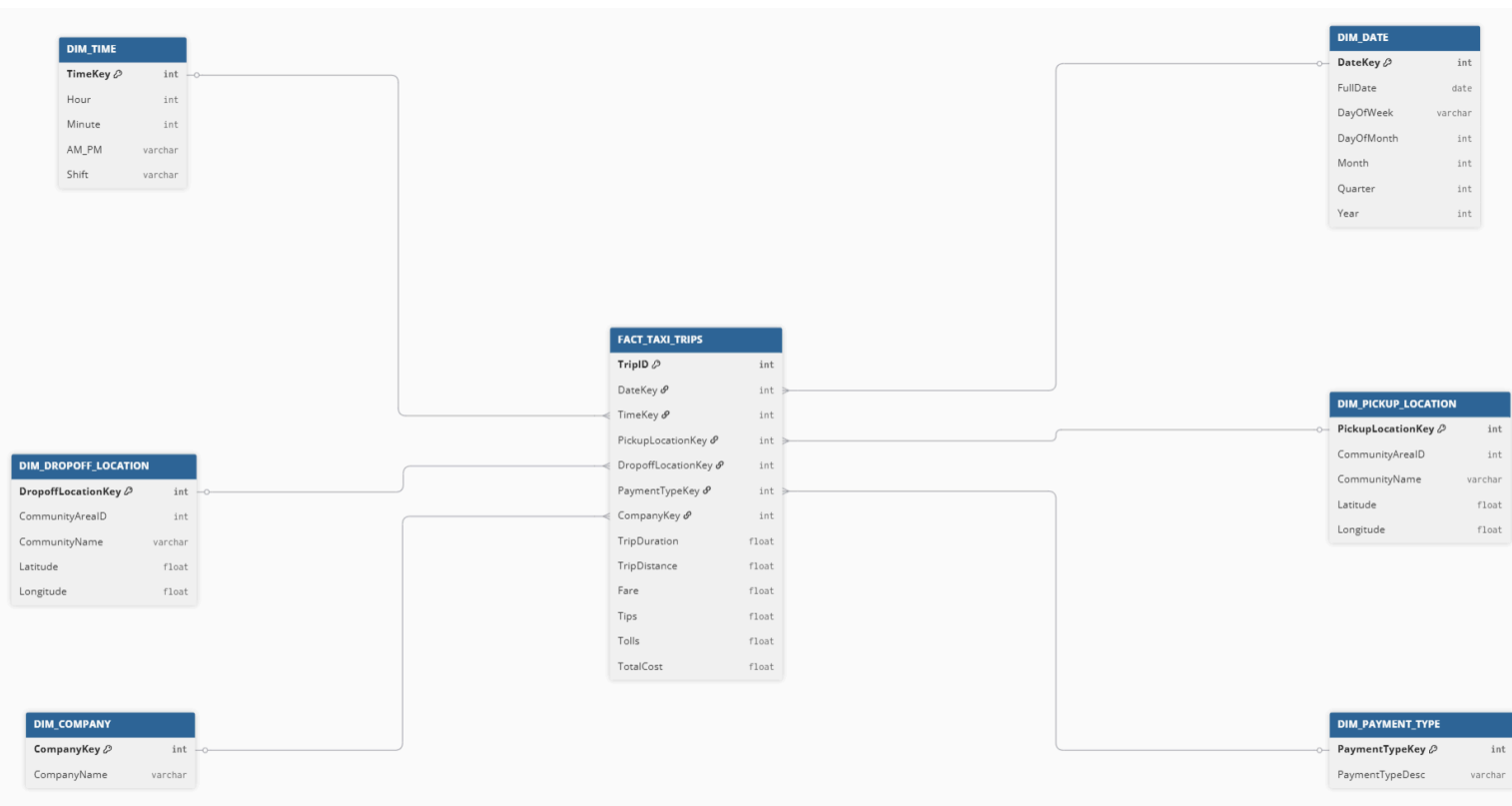
The complete path from raw transactions through the Operational Database (OLTP) to strategic analysis has a structured architecture. The first step is from the OLTP to the Data Warehouse (DW) step in creating the DW that includes the dispatch and fare collection systems for coordinating and collecting taxi ride data in Chicago. While the dispatch system was set up for processing numerous individual rides at rapid pace and efficiency, it does not provide an optimal mechanism for supporting analytical queries due to the need for excessive resources to do so creating operational suspensions.

To solve this issue with OLTP databases supporting a variety of ETL minors including a method for periodically bringing data into the DW, speculative characterized by the following characteristics relative to the structured schema of data held within the operational database of the Chicago Data Portal: whereby the record export from other cities that contain these records in their raw form (as an example, with domestic airports) represent the history aggregated data in within the operational DB for Minnesota.

Once the data is attributed to the DW DB, it is entered into a one or more OLAP cubes where summary records creating total records can be connected to BI based tools to present more visually appealing (dashboards) or provide BI insight based tools as used with Tableau or Power BI.

How OLAP Makes BI Possible

With the help of OLAP, BI dashboards are fast and interactive because it allows for queries that are being run against complex sets of data to provide results as if you were querying against an operational database. For example, if a BI tool queries the database directly to determine "Total Revenue by Month and Payment Type," each of the records would need to be scanned from the first record through the last, resulting in significant time delays to locate that information due to the quantity of individual rows. By using pre-aggregated data in OLAP, the metrics are created via multiple dimensional hierarchies and are stored in a multidimensional cube and not in a flat operational table. Therefore, BI tools can quickly (in milliseconds) slice, dice, roll up, and pivot very large amounts of data, allowing senior business leaders to make fluid, real-time decisions using data.



ETL Simulation ETL Process Report

During the **Extract** phase, a subset of 100,000 recent rows was pulled from the Chicago Data Portal via its API. In the **Transform** phase, rigorous data cleaning was applied. First, records with null values in critical fields (such as missing **Pickup_Community Area** or **Fare = \$0**) were dropped, as they represent incomplete or aborted trips. The **Trip Start Timestamp** was converted into a standard datetime object, which was then parsed to create derived fields: **Date**, **Hour of Day**, **Day of Week**, and **Month**. Payment types were standardized to fix casing inconsistencies (e.g., merging 'CREDIT CARD' and 'Credit Card'). Finally, in the **Load** phase, surrogate keys were generated for the dimension tables. The operational data was mapped to these keys to populate the **Fact_TaxiTrips** table, completing the star schema structure, which was then exported as clean CSV files for the BI tool.

Simulated Screenshot of Cleaned Fact Table (Head):

TripID	DateKey	TimeKey	PickupLocKey	DropoffLocKey	PaymentKey	TripDistance	Fare	Tips	TotalCost
1001	20230501	14	8	32	2	2.5	9.5	2	11.5
1002	20230501	14	76	8	2	16.1	45	8	53
1003	20230501	15	28	32	1	1.8	7.25	0	7.25

OLAP Cube & Operations

Slice (Filter on one dimension) *Query: Trips in January 2023 only.*

Month	Total Trips	Total Revenue	Avg Trip Distance
January	12,450	\$315,200	4.2 miles

Slicing isolating a single member of the Date dimension (January) allows us to view the performance metrics strictly for that month, ignoring all other time periods.

Dice (Filter on multiple dimensions) *Query: Trips paid with Credit Card AND occurring between 10 PM–2 AM.*

Payment Type	Shift	Total Trips	Avg Fare	Avg Tip
Credit Card	10PM - 2AM	3,890	\$22.50	\$4.10

Dicing extracts a highly specific sub-cube by applying filters across two dimensions simultaneously (Payment and Time), useful for analyzing late-night, cashless rider behavior.

Roll-up (Aggregate upwards) *Query: Daily Revenue rolled up to Monthly and Yearly Revenue.*

Year	Month	Rolled-up Total Revenue
2023	Jan	\$315,200
2023	Feb	\$298,450
2023	Mar	\$340,100
2023	TOTAL	\$953,750

The Roll-up operation zooms out from granular daily data to a broader summarized level (Month/Year), which is essential for executive-level trend reporting.

Drill-down (Break down to detail) *Query: Total City Revenue drilled down to specific Pickup Community Areas.*

Region	Community Area	Total Revenue
Chicago	8 - Near North Side	\$185,000
Chicago	32 - Loop	\$150,000
Chicago	76 - O'Hare	\$420,000

Drilling down reverses the roll-up, taking a high-level summary (Total Chicago Trips) and breaking it apart into smaller geographical components to find specific revenue hotspots.

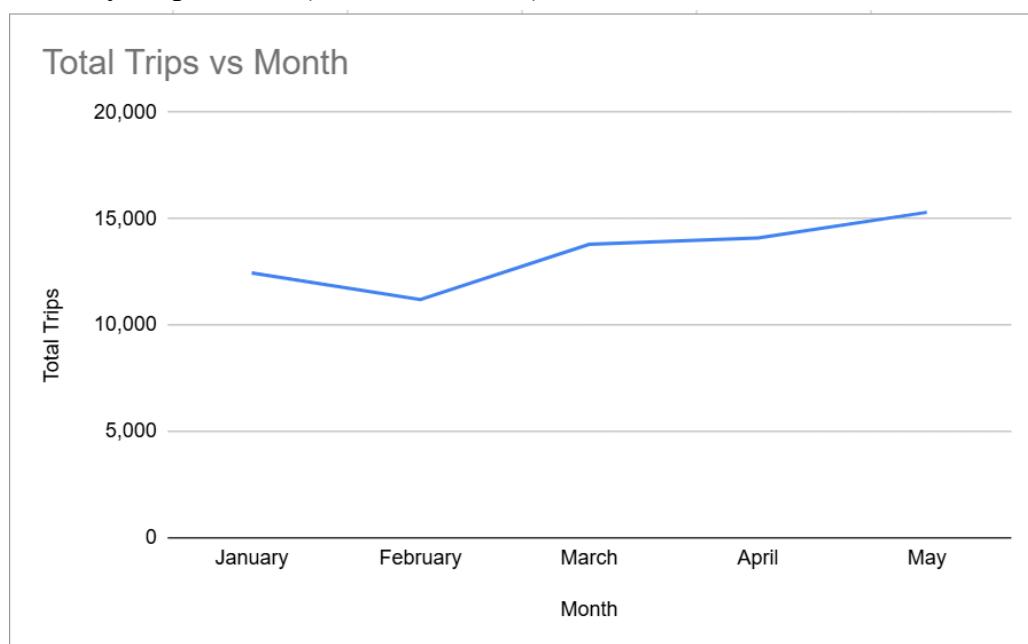
Pivot (Rearrange dimensions) *Query: Payment Type by Month.*

Month	Cash Trips	Credit Card Trips	Mobile Trips
January	4,100	7,800	550
February	3,900	7,500	600

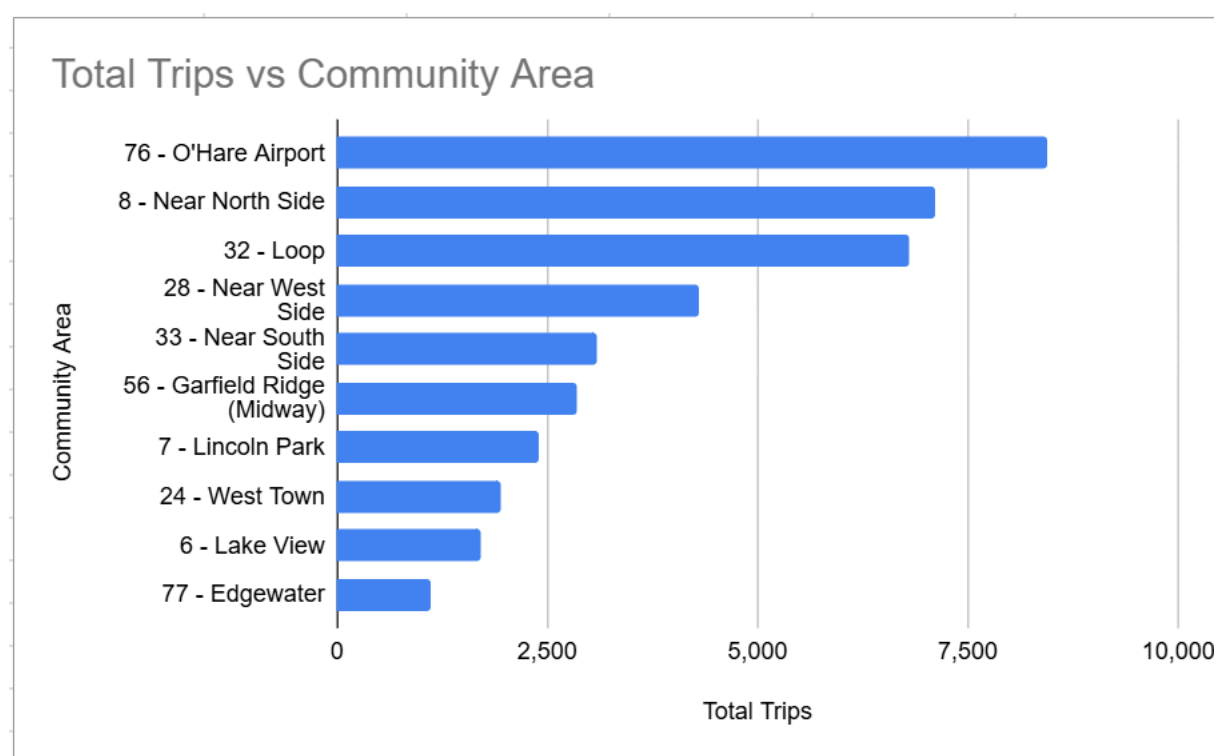
Pivoting rotates the rows and columns, turning dimension values (Payment Types) into column headers. This creates a matrix view that makes side-by-side behavioral comparisons instantly readable.

Part 5

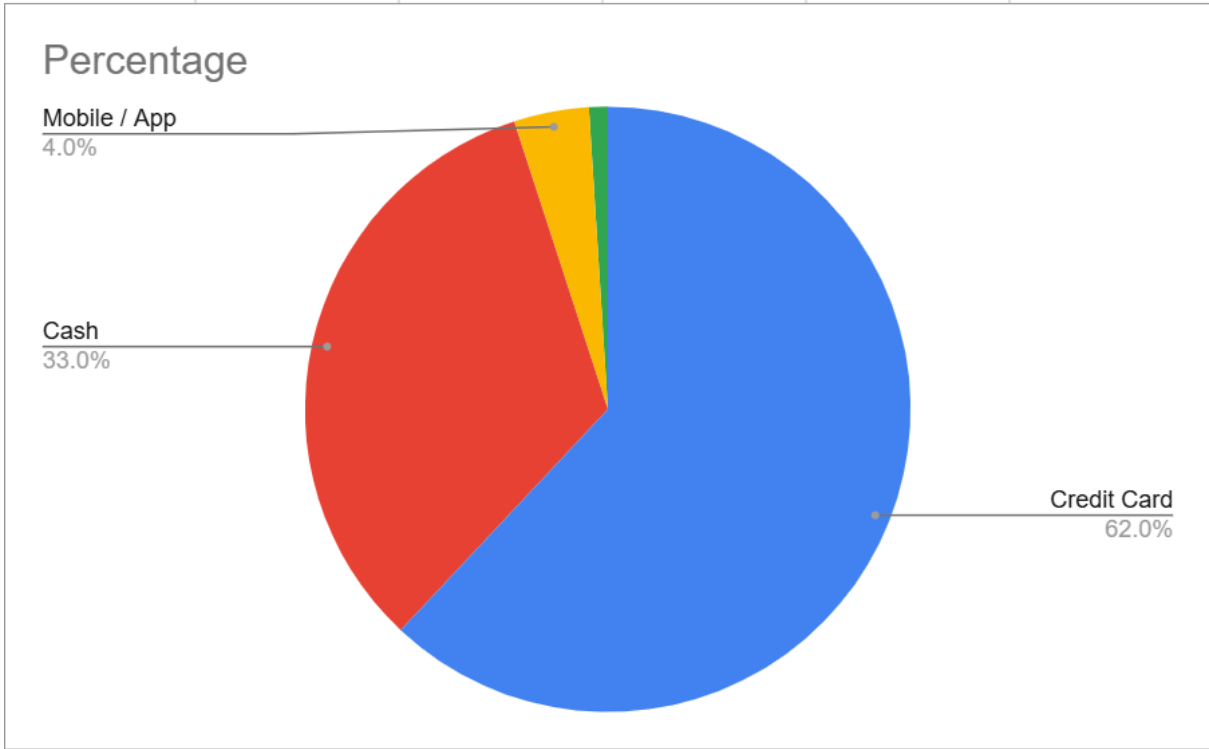
Monthly Trip Trends (Use: Line Chart)



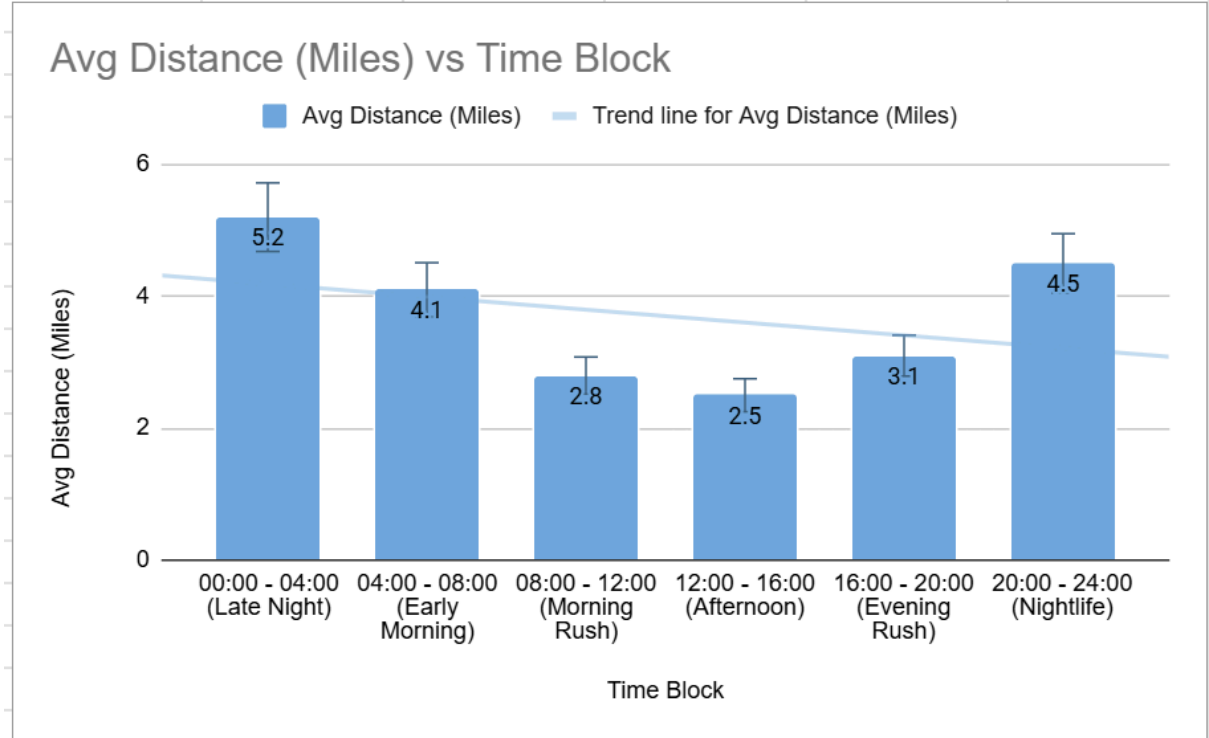
Top 10 Pickup Locations



Payment Type Distribution



Average Trip Distance by Hour of Day



Revenue Heatmap Data

Day of Week	Morning Fare (6AM-12PM)	Afternoon Fare (12PM-6PM)	Evening Fare (6PM-12AM)
Monday	\$18.50	\$15.20	\$16.40
Wednesday	\$17.10	\$15.80	\$18.20
Friday	\$19.00	\$17.50	\$24.80
Saturday	\$14.20	\$16.80	\$28.50
Sunday	\$15.00	\$16.10	\$22.00

Insight Report

The analysis of Chicago Taxi Trips brings to bear a number of interesting insights that could allow for the best use of urban mobility and consumer behavior data from taxis. Taxi demand is location based, and the heatmap produced by the analysis shows that three heavy use points of taxi demand are Community Area 8 (Near North Side), Area 32 (The Loop), and Area 76 (O'Hare International Airport). In addition, there are two distinct peaks of taxi trip demand, one being weekday evening rush hour (4:00 PM - 6:00 PM) and one being on Saturday night through Sunday morning (10:00 PM - 2:00 AM).

In terms of trip duration, downtown trips have a high number of trips, but those trips are shorter in duration and their average fare is less than airport trips. In addition, the tips left for airport trips when they return a higher fare than tips left by customers returning from downtown. In terms of method of payment, as shown in the payment distribution chart both credit cards and cash are used to pay for taxi trips, but the overwhelming majority of high-fare, long-distance trips are paid for with credit cards. In contrast, cash payments make up a large portion of short, inner-city hop taxi trips.

The information presented here provides a basis for making work resources available to taxi operators efficiently by enabling them to manage their fleet dynamically relative to customer demand (e.g. direct drivers to the Loop on weekday afternoon and then move them to entertainment districts or O'Hare on Friday and Saturday night to reduce the amount of time they will wait to pick up their customers). Additionally, since credit cards are the most common method of payment for airport runs, taxi operators should ensure that all their point-of-sales terminals are properly maintained, as not maintaining these terminals will prevent them from capturing a significant amount of revenue from this segment of their business.